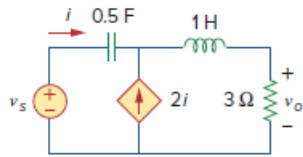


Problem

Obtain the transfer function $H(s) = V_o / V_s$ for the circuit of Fig. 16.96.

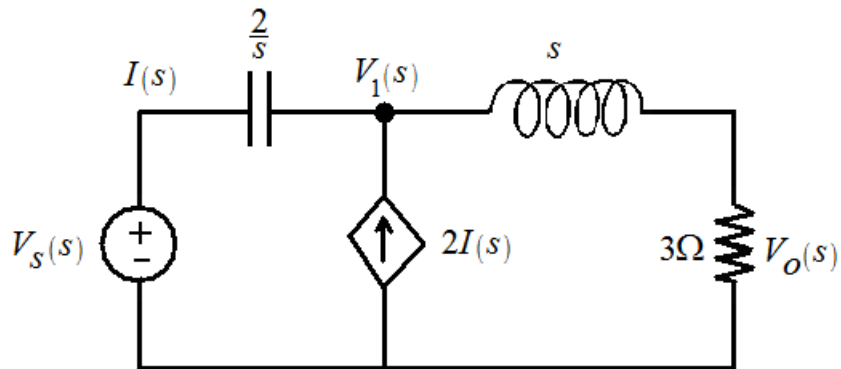
Figure 16.96:



Step-by-step solution

Step 1 of 4

Given circuit in complex domain



Step 2 of 4

At node 1 we can write

$$2I(s) + I(s) = \frac{V_1(s)}{s+3}$$

$$3I(s) = \frac{V_1(s)}{s+3} \dots \dots \dots (1)$$

From the circuit we can write

$$I(s) = \frac{V_s(s) - V_1(s)}{\frac{2}{s}}$$

Step 3 of 4

Now equation (1) becomes

$$3 \left(\frac{V_s(s) - V_1(s)}{\frac{2}{s}} \right) = \frac{V_1(s)}{s+3}$$

$$V_s(s) \left(\frac{3s}{2} \right) = V_1(s) \left(\frac{1}{s+3} + \frac{3s}{2} \right)$$

$$V_s(s) \left(\frac{3s}{2} \right) = V_1(s) \left(\frac{2+3s(s+3)}{2(s+3)} \right)$$

$$V_1(s) = V_s(s) \left(\frac{3s(s+3)}{2+3s(s+3)} \right)$$

$$V_1(s) = V_s(s) \left(\frac{3s(s+3)}{3s^2+9s+2} \right) \dots\dots\dots (2)$$

Step 4 of 4

From the circuit we can write

$$V_o(s) = V_1(s) \left[\frac{3}{3+s} \right]$$

From equation (2)

$$V_o(s) = V_s(s) \left(\frac{3s(s+3)}{3s^2+9s+2} \right) \left[\frac{3}{3+s} \right]$$

$$V_o(s) = V_s(s) \left(\frac{9s}{3s^2+9s+2} \right)$$

$$\boxed{\frac{V_o(s)}{V_s(s)} = \left(\frac{9s}{3s^2+9s+2} \right)}$$